

Web Development Notes (Basic to Advanced)

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Web Development is the process of creating websites and web applications. It includes Frontend, Backend, Databases, APIs, Security, Deployment and Cloud technologies.

HTML

HTML is the standard markup language for creating web pages. Important tags include headings, paragraphs, links, images, forms, tables, divs and semantic tags like header, footer, section and article.

CSS

CSS is used to style web pages. Topics include selectors, box model, margins, padding, flexbox, grid layouts, responsive design and media queries.

JavaScript

JavaScript adds interactivity to websites. Important concepts include variables, data types, functions, arrays, objects, loops, conditions, DOM manipulation, events, fetch API, async/await and ES6 features.

Bootstrap

Bootstrap is a CSS framework that provides prebuilt responsive components such as navbars, cards, buttons, forms and grid systems.

Backend Development

Backend development handles business logic, authentication, APIs and database communication using technologies such as PHP, Python Flask, Java and Node.js.

Flask Framework

Flask is a lightweight Python framework. Core concepts include routes, templates, forms, sessions, blueprints and database integration.

Database

Databases store application data. SQL operations include CREATE, INSERT, SELECT, UPDATE and DELETE. Popular databases include MySQL, PostgreSQL and TiDB.

Authentication

Authentication verifies user identity through signup, login, password hashing, sessions, JWT tokens and OTP verification.

REST APIs

REST APIs enable communication between applications using GET, POST, PUT and DELETE methods with JSON data.

Git & GitHub

Git is a version control system used to track changes. Common commands are git init, add, commit, push, pull and clone.

Deployment

Deployment publishes a web application online using services such as Render, Railway, VPS servers and cloud platforms.

Web Security

Important security topics include SQL Injection prevention, XSS protection, CSRF protection, HTTPS and secure password storage.

Advanced Topics

Advanced web development includes WebSockets, Docker, Cloud Computing, Microservices, Caching, CI/CD and Kubernetes.

Roadmap

HTML → CSS → JavaScript → Bootstrap → Backend (Flask/PHP) → Database → Authentication → APIs → Security → Deployment → Cloud.